

A Deterministic Referral-Safety Contract for Medical-LLM Extraction of Time-Critical Otologic Symptoms: An Open Benchmark Evaluation

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Abstract

Objective: To test a contract-based architecture that separates a non-overridable referral rule from the fallible text extraction that feeds it. **Materials and Methods:** Seven guideline-informed rules were evaluated on 71 author-curated cases (34 structured, 37 free-text; no patient data). We measured rule coverage on encoded features and end-to-end performance through naive and benchmark-tuned rule references plus `google/medgemma-4b-it`, using a shared schema, adapter, and metrics. **Results:** On the structured cases, the layer reached 100% sensitivity (19/19; 95% CI, 83–100) and specificity (15/15; 80–100). With identical free text, recall was 56% (10/18; 34–75) for the naive reference, 78% (14/18; 55–91) for MedGemma-4B, and 100% (18/18; 82–100) for the in-sample tuned reference. MedGemma specificity was 100% (19/19; 83–100); misses concentrated in colloquial and ambiguous phrasing. **Discussion:** The architecture ensures that a fired rule cannot be suppressed, but it cannot rescue a feature the extractor misses. Author-curated cases, one independently developed LLM, and small samples limit inference. **Conclusion:** Separating the software invariant from extraction performance localizes residual risk and defines a protocol for validation on adjudicated clinical notes.

Keywords: clinical NLP; large language models; patient safety; referral triage; reproducibility.

1 Background and Significance

Sudden sensorineural hearing loss (SSNHL) is an otologic emergency: prompt audiometric confirmation and early corticosteroid therapy—within roughly two weeks—offer the best chance of recovery, and delayed presentation is associated with worse outcomes [1]. In an era of LLM-assisted clinical documentation and triage, a foreseeable failure mode is that a probabilistic model, extracting or summarizing a patient narrative, assigns a reassuring “low-risk / likely stress” interpretation to what is in fact an acute asymmetric loss—and thereby contributes to delay. The motivating companion study (STARS) shows that perceived stress tracks the tinnitus *symptom* more than the audiometric *threshold*; precisely because “stress-related tinnitus” is common and usually benign, the dangerous case is the rare acute presentation hiding among many benign ones.

The safe design principle is therefore not merely a better classifier but a *non-overridable floor*: a deterministic, guideline-informed rule layer that screens for red flags and forces urgent referral regardless of any model score. This paper treats that floor as a testable *safety contract*, not as an informal guardrail claim. The contract has a narrow invariant—once an affirmative red-flag feature reaches the layer, no downstream model score can reverse the urgent disposition—and an explicit assumption—the extractor must surface the feature. Separating the invariant from its assumption makes residual risk measurable rather than hidden inside an end-to-end accuracy number.

Research questions. We ask three evaluation questions. **RQ1 (coverage):** given encoded structured features, does the rule layer cover the benchmark red-flag set without flagging benign controls? **RQ2 (propagation):** how much performance is lost when features must first be extracted from text? **RQ3 (independence):** does the extraction bottleneck persist for an independently developed medical LLM that did not contribute benchmark cues or rule logic? These questions distinguish software verification from validation of the complete clinical pipeline.

Contributions.

1. A **contract-based decomposition** of referral safety into a verifiable deterministic invariant and a measurable extraction assumption. This identifies exactly what the safety layer can guarantee and what it cannot.

2. A guideline-informed rule set and **two-level benchmark** (71 author-curated cases; no patient data) that evaluates rule coverage separately from text-to-disposition performance, preventing perfect rule execution from being mistaken for complete pipeline safety.
3. A controlled **three-system stress test** on identical notes: naive rules, an explicitly in-sample tuned upper reference, and an independently developed open medical LLM. The observed 56%–78%–100% recall range localizes the dominant failure mechanism to extraction and context handling rather than the final OR logic.
4. A reproducible research artifact—typed schema, model-neutral adapter, deterministic rules, case benchmark, metrics, unit tests, and open-LLM runner—that lets future studies replace the extractor or notes while holding the safety contract fixed.

1.1 Related work

Clinical information extraction with LLMs has advanced rapidly: general-purpose LLMs act as strong few-shot clinical information extractors [3], building on a decade of structured clinical-NLP benchmarks for concepts, assertions, and relations in clinical text [4], and medical-domain models now encode substantial clinical knowledge [5]. Open medical models such as MedGemma are positioned as starting points for health-AI development rather than validated devices [6, 7]. Capability, however, is not safety: a widely deployed proprietary sepsis model performed poorly on external validation [8], and dataset shift routinely degrades clinical AI in ways a clinician must be positioned to catch [9]—which is why reporting standards for clinical prediction/AI emphasize prespecified, transparent evaluation [10]. A recurring theme in safety-critical clinical AI is therefore that probabilistic components should be wrapped in deterministic guardrails for time-critical decisions; our contribution is a concrete, guideline-traceable instance for otology, released *with* an open benchmark and honest end-to-end numbers rather than an asserted target. The clinical endpoints follow the SSNHL and tinnitus guidelines [1, 2]. This work develops the safety/extraction component of the STARS program with results; the empirical survey findings are reported separately.

2 Materials and Methods

2.1 System architecture: extraction, then a non-overridable safety floor

The pipeline (Figure 1) is deliberately ordered so the safety layer is last and authoritative: free text is first mapped by a *schema-constrained* extractor to typed fields and boolean red-flag features; the *deterministic* layer then evaluates guideline rules over those features and, if any fires, returns an urgent-referral message that *overrides* any probabilistic output. A low model score can never suppress this message.

Formally, let $E(t) \in \{0, 1\}^k$ be the feature vector extracted from note t , let r_j be the j th guideline rule, and define the safety disposition

$$S(t) = \bigvee_{j=1}^7 r_j(E(t)).$$

The architectural invariant is *monotone override*: if any $r_j(E(t)) = 1$, the emitted disposition is urgent irrespective of a probabilistic score $p(t)$. This is a software property of composition order, not a claim that $E(t)$ is clinically complete. End-to-end false negatives occur when the clinically relevant feature is absent from or misrepresented in $E(t)$; the two-level evaluation measures these two failure surfaces separately.

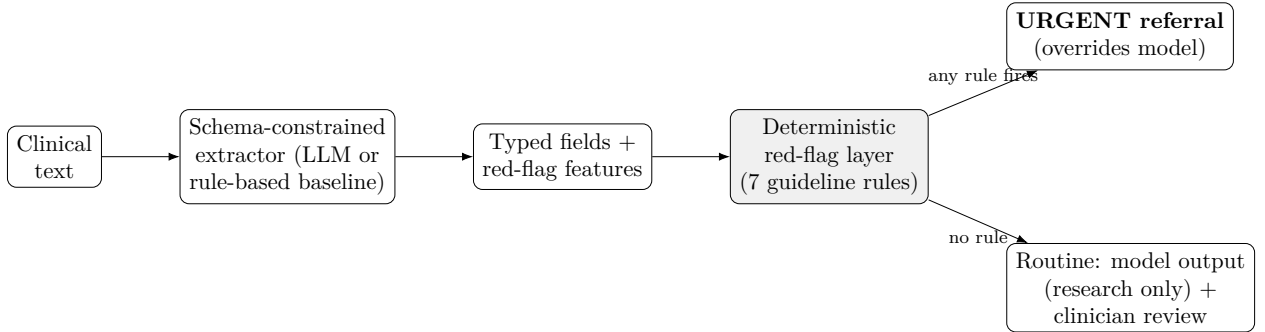


Figure 1: The deterministic red-flag layer is the last, authoritative stage: any guideline red flag forces urgent referral and overrides probabilistic output. The extractor may be an open LLM (e.g., MedGemma) or the reproducible rule-based baseline evaluated here.

2.2 Guideline-informed operational rules

Seven operational triggers were informed by the SSNHL and tinnitus guidelines [1, 2]: (1) sudden hearing loss; (2) rapidly progressive loss; (3) unilateral sudden symptom; (4) acute aural fullness with a sudden drop; (5) single-sided tinnitus *with* audiometric asymmetry; (6) vertigo; and (7) a focal neurologic sign (Table 1). The decision is a logical OR; the tinnitus/asymmetry rule requires both indicators. These are conservative benchmark operationalizations, not verbatim key action statements, and the prototype’s “urgent” output has not been validated against clinician disposition or outcomes. Determinism makes implementation behavior auditable; it does not establish clinical completeness or appropriateness.

Table 1: **Operational rule specification and clinical rationale.** These are conservative research-benchmark triggers informed by the cited guidelines; they are not verbatim guideline recommendations or a validated triage protocol.

Encoded trigger	Intended concern	Prototype action
Sudden hearing loss	Time-sensitive assessment for possible SSNHL	Urgent evaluation
Rapidly progressive loss	Short-course deterioration requiring prompt assessment	Urgent evaluation
Unilateral sudden symptom	Acute asymmetric presentation	Urgent evaluation
Acute fullness with sudden drop	Possible acute hearing change despite nonspecific fullness	Urgent evaluation
Single-sided tinnitus <i>and</i> audiometric asymmetry	Combined localizing symptom and objective asymmetry	Urgent evaluation
Vertigo	Conservative vestibulo-otologic warning trigger	Urgent evaluation
Focal neurologic sign	Possible central neurologic process	Urgent evaluation

2.3 Intended use and governance requirements

The evaluated system is a research prototype, not a diagnostic or autonomous triage device. The extractor is prompted to return a fixed typed schema rather than a disposition, and the rule layer operates only on that schema. The present benchmark contains no clinicians or workflow evaluation; therefore clinician verification, provenance capture, monitoring, escalation on parse failure, and prospective audit are *requirements for future evaluation*, not components validated here. A finite-set sensitivity of 1.00 does not imply zero deployment misses.

2.4 Open benchmark construction (no patient data)

The single author curated all cases from the encoded rule concepts and plausible linguistic variants; none are clinical records. The benchmark can therefore be released openly but is not an external clinical validation set and has no independent clinical adjudication. It contains 71 cases: 34 *structured* cases (19 urgent by the benchmark definition, 15 benign controls) and 37 *free-text* notes (18 urgent, 19 benign). Text cases span four deliberately constructed challenge categories—*typical* (clean phrasing), *negation* (explicit negatives), *colloquial* (lay phrasing), and *ambiguous* (vague onset/laterality)—and include noncanonical urgent wording and resolved historical events containing red-flag terms. Labels encode the intended rule semantics before an extractor is applied, but benchmark authorship, labels, and rule design are not independent. Each case carries laterality, onset, and vertigo/neurologic metadata for descriptive subgroup analysis.

2.5 Two-level evaluation

We evaluate at two levels. **Rule coverage** applies the deterministic layer to encoded structured features, testing consistency between the benchmark definition and its implementation. **End-to-end** runs each free-text note through an extractor and then the same safety layer, allowing extraction error to propagate. Red-flag recall (sensitivity) is the primary descriptive metric; we also report specificity and over-referral with 95% Wilson confidence intervals, plus field-level and note-category performance. No hypothesis test or model-superiority claim was planned because only 18 text cases are urgent. Categories are challenge strata, not demographic or clinical subgroups.

2.6 Three-system extractor comparison

We compare three deliberately contrasting systems on the same 37 notes: (i) a naive keyword reference (v1), (ii) an improved deterministic reference (v2) adding historical/resolved-context suppression, colloquial and implicit-laterality cues, and keyword co-occurrence, and (iii) the independently developed medical LLM `google/medgemma-4b-it`. The first two bracket transparent under- and over-tuning behavior; the third tests whether the extraction bottleneck survives a change in model family and authorship. Neither rule system is presented as state of the art, and v2 is explicitly an *in-sample, benchmark-tuned upper reference*, not an external-validation result.

The open-LLM path is implemented, not hypothetical: `llm_medgemma.py` loads the 4-billion-parameter instruction-tuned checkpoint through `transformers`, uses a single zero-shot prompt, greedy decoding (temperature effectively zero), and at most 256 newly generated tokens, constrains output to the fixed extraction schema, and parses JSON robustly (code-fence/prose tolerant, brace-balanced, with a conservative all-*unknown* fallback that can only *under-flag*). No examples, benchmark cue lists, or gold labels are included in the prompt, and there is no model-specific post-processing. A schema-to-feature adapter maps all three systems to the same boolean red-flag features, after which `run_llm_eval.py` applies one identical metric implementation. We report per-feature precision/recall/F1, document-level exact feature match, urgency-changing error rate (an extraction error that flips the layer’s decision), run-to-run consistency, and end-to-end sensitivity/specificity with Wilson intervals. The released registry also supports two ungated compact controls (Qwen2.5-0.5B-Instruct and SmolLM2-360M-Instruct); these are executable extensions and are not included in the reported results because they were not part of the frozen three-system run.

3 Results

3.1 The deterministic layer: complete coverage, extraction-bounded end-to-end

RQ1—coverage. On correctly-extracted features the layer flagged every benchmark red flag and over-referred none: sensitivity 100% (19/19) and specificity 100% (15/15) (Table 2). This verifies coverage of the *encoded benchmark taxonomy*—not all possible clinical emergencies—and shows that the combined single-sided-tinnitus-with-asymmetry rule leaves the prespecified benign single-sided-tinnitus control unflagged.

RQ2—error propagation. End-to-end performance was dominated by extraction. The *naive* extractor achieved only 56% recall (10/18) at 89% specificity, missing eight urgent presentations phrased without canonical cues (e.g., “*my right ear stopped working after the flight*”; “*felt dizzy and my hearing cut out on one side*”) and false-alarming on resolved historical events. The *improved* extractor reached 100% recall and 100% specificity *on this benchmark* (Table 2), recovering the colloquial and ambiguous cases and suppressing historical false alarms (recall by category rose from 43% to 100% for colloquial and 33% to 100% for ambiguous notes; Table 3). Thus the same invariant layer ranges from 56% to 100% recall solely because its inputs change. The finite-set 100% is an

in-sample ceiling, not a deployment guarantee.

Table 2: **Deterministic referral-safety layer on the open benchmark** (34 structured + 37 text cases; author-curated/synthetic, no patient data). Sensitivity is red-flag recall (the safety-critical metric); 95% Wilson CIs. *Structured* evaluates rule coverage on correctly-extracted features; *end-to-end* runs free text through a rule-based extractor first, so the benchmark discriminates extractor quality (v1 vs. the improved v2). Generated by `code/src/run_redflag_eval.py`.

Evaluation	Pos.	Neg.	Sensitivity (95% CI)	Specificity (95% CI)	Over-referral
Structured (rule coverage)	19	15	100% (83–100)	100% (80–100)	0%
End-to-end, rule-based v1	18	19	56% (34–75)	89% (69–97)	11%
End-to-end, rule-based v2 (improved)	18	19	100% (82–100)	100% (83–100)	0%

Table 3: **End-to-end red-flag recall by note category and extractor.** Adversarial phrasing (colloquial, ambiguous, historical) is where extraction error — not the deterministic rules — creates residual risk; the improved extractor (v2) recovers most of it, quantifying the benchmark’s discriminative value. ‘–’ = no positive case in that cell. Generated by `code/src/run_redflag_eval.py`.

Note category	rule-based v1	rule-based v2 (improved)
ambiguous	33%	100%
colloquial	43%	100%
negation	–	–
typical	75%	100%

3.2 Extraction baseline

Field-level metrics track the end-to-end gap (Table 5): the naive extractor reached macro-F1 0.72 with a 27% urgency-changing error rate, while the improved extractor reached macro-F1 0.94 with a 0% urgency-changing error rate on this benchmark; both were perfectly reproducible (run-to-run consistency 1.00). Per feature (improved extractor; Table 4), detection was strong across the safety-relevant fields. Two structural gaps we previously flagged—the extraction *schema* not encoding audiometric *asymmetry* or a *rapidly-progressive* course—have since been closed: the schema now carries an `audiometric_asymmetry` field and a `rapidly_progressive` course value, and the adapter maps both to their red-flag features (verified by unit tests), so an LLM mapped through the adapter no longer inherits those blind spots. The rule-based baseline numbers are unaffected (the adapter feeds only the LLM path).

Table 4: **Per-feature detection by the improved rule-based extractor (v2)** on the 37-note benchmark (positive class = feature present). The improved extractor is an in-sample, benchmark-tuned reference rather than an external comparator. Generated by `code/src/run_extraction_eval.py`.

Feature	Prec.	Rec.	F1	Support
Sudden hearing loss	0.82	1.00	0.90	9
Rapidly progressive loss	0.80	1.00	0.89	4
Unilateral acute symptom	1.00	0.93	0.97	15
Acute aural fullness	1.00	1.00	1.00	4
Single-sided tinnitus	1.00	0.60	0.75	5
Asymmetric hearing	1.00	1.00	1.00	2
Vertigo	1.00	1.00	1.00	4
Focal neurologic sign	1.00	1.00	1.00	1
Macro average	0.95	0.94	0.94	

3.3 Open medical-LLM (MedGemma) evaluation

RQ3—*independent extractor*. The open-LLM comparison is not a promise but a measurement. A single command scores every extractor—the two rule-based references and an actual open medical LLM, `google/medgemma-4b-it`—through the identical benchmark, adapter, and metrics (we ran MedGemma on a free-tier GPU with greedy, deterministic decoding):

```
python src/run_llm_eval.py -extractor medgemma \
    -model google/medgemma-4b-it \
    -out ../paper02/outputs/llm_extractor_eval.json
```

Table 5: **Extractor comparison through the identical safety harness.** MedGemma used the same author-curated benchmark and schema adapter as the rule references but did not receive their cue lists. The tuned rule extractor is an in-sample reference, and this descriptive comparison is not an external validation or powered model ranking. Generated by `code/src/run_llm_eval.py`.

Extractor	Macro F1	Doc. exact	Urg.-chg. err.	E2E recall	E2E spec.	Run consist.
Rule-based v1	0.72	0.59	0.27	56%	89%	1.00
Rule-based v2 (improved)	0.94	0.84	0.00	100%	100%	1.00
MedGemma (open LLM)	0.75	0.57	0.11	78%	100%	1.00

Through this harness, MedGemma reached **78% end-to-end red-flag recall** (14/18; 95% Wilson CI 55–91%) at **100% specificity** (19/19; over-referral 0%), with field-level macro-F1 0.75 and a 10.8% urgency-changing error rate; greedy decoding made it perfectly reproducible (run-to-run consistency 1.00). It therefore sits *between* the two rule-based references—naive 56%, MedGemma 78%, tuned 100%—on identical cases (Table 5). Its four missed urgent notes were the implicitly-

phrased ones: recall was 88% on typical phrasing but fell to 71% (colloquial) and 67% (ambiguous), the same categories where the naive rule extractor failed. Notably, although MedGemma *over-*asserted audiometric asymmetry at the field level (precision 0.13, liberally reading “asymmetry” from narrative text), this did not produce a single over-referral: the conservative combined rule—single-sided tinnitus *with* asymmetry—absorbed the over-call, a direct validation of the combined-rule design. The plumbing itself is separately guaranteed by seven unit tests that exercise the full path (prompt construction, generation, code-fence/prose-tolerant JSON parsing, schema-to-feature mapping, safety layer, metrics) with a deterministic no-model stand-in.

This is a descriptive small-sample comparison, not a model-ranking exercise. The sensitivity point estimates differ by 22 percentage points between each adjacent system, but the corresponding Wilson intervals are wide and paired significance testing would be underpowered with only 18 urgent notes. We therefore report numerators, denominators, and intervals rather than claiming superiority. The clinically relevant finding is qualitative and consistent across field- and decision-level metrics: changing the extractor changes whether urgent cases reach the non-overridable layer.

MedGemma did not author the benchmark or receive its cue lists, so its result is independent of the two rule-extractor implementations. It is not independent of the author-defined notes, labels, schema, or adapter. The result therefore provides a useful implementation-level stress test, not external validation and not a remedy for all circularity. Its four misses show that the benchmark is not saturated by this checkpoint; they do not establish an expected deployment miss rate. Reproduction requires the released code, the specified model checkpoint and software environment, and access to compatible compute.

4 Discussion

The central contribution is a clean separation of claims. The deterministic layer is *fully auditable and invariant given its inputs*: within the encoded benchmark it achieved complete rule coverage and zero over-referral, and composition order ensures that a fired rule cannot be suppressed by a low probabilistic score. This is narrower—and more defensible—than claiming that a guardrail makes the complete pipeline safe. End-to-end safety is governed by extraction: red-flag recall was 56% (naive extractor), 78% for MedGemma-4B, and 100% (in-sample tuned reference) on identical cases.

A red flag the extractor never surfaces cannot be rescued downstream, while mishandled historical context can create false alarms. The same qualitative failure axes—colloquial phrasing, implicit laterality, and ambiguous temporal context—appeared across the naive rules and independent LLM, supporting extraction as the mechanism rather than an artifact of one implementation.

The broader informatics value is methodological. Reporting only rule-level performance would have yielded an apparently perfect system; reporting only end-to-end performance would not identify whether errors came from clinical logic or language understanding. The contract-based decomposition makes both visible and creates a stable interface for substitution experiments: an institution can hold rules and metrics fixed while replacing the extractor, or hold the extractor fixed while extending the red-flag taxonomy. This pattern is applicable to other time-critical pathways whenever a finite, auditable override can be separated from probabilistic interpretation, although the otology-specific rules and observed performance do not transfer without revalidation.

The practical implications are specific. First, extraction is the observed bottleneck *within this benchmark*: holding the final rule layer fixed produced recall from 56% to 100%. Second, the tuned extractor’s finite-set 100% is not evidence of transportability. Third, MedGemma’s 78% recall would have routed 4 of 18 benchmark-urgent notes to the nonurgent path if used without review. This count motivates clinician verification and parse-failure escalation in a future pilot, but it must not be extrapolated to clinical prevalence or harm without real-note evaluation.

4.1 Limitations

This is a small, author-curated benchmark, so confidence intervals are wide and challenge-category counts are sparse. The same author specified the rules, authored or curated the cases, assigned labels, and tuned v2; there was no independent clinical review or inter-annotator agreement assessment. Structured “coverage” primarily tests implementation consistency. The MedGemma evaluation uses one 4B instruction-tuned checkpoint, one zero-shot prompt, deterministic decoding, and author-defined schema mapping. Results may change with checkpoint revision, prompt, quantization, software version, or hardware. No patient notes, demographic groups, workflow outcomes, time-to-referral outcomes, calibration, comparative clinician performance, or prevalence-weighted alert burden were evaluated. Cases are English-language and deliberately enriched for red flags; predictive values and workload cannot be inferred. These limitations preclude clinical-effectiveness, fairness,

transportability, or deployment claims.

4.2 Threats to validity and next validation step

Construct validity: guideline traceability does not establish that seven encoded rules exhaust all urgent otologic presentations, and synthetic notes cannot reproduce the full ambiguity of clinical documentation. **Internal validity:** v2 was refined against this benchmark and is therefore an in-sample upper reference; only MedGemma supplies an authorship-independent extractor check. **External validity:** one English benchmark, one medical-LLM checkpoint, and no patient notes preclude estimates of transportability, prevalence-dependent workload, or clinical utility. **Statistical conclusion validity:** 18 urgent notes support uncertainty intervals and failure discovery, not powered model ranking.

The next decisive study is therefore not another tuned synthetic score but a locked, prospective pilot on consecutively sampled, deidentified otology notes. Before access, the schema, rules, prompt, and primary endpoint should be frozen; two clinicians should independently annotate red-flag features and disposition with adjudication; all eligible notes should be retained; and sensitivity, specificity, abstention/parse-failure rate, review time, and disagreement should be reported with confidence intervals. The present benchmark supplies the executable protocol and failure taxonomy for that study but does not substitute for it.

5 Conclusion

A deterministic, guideline-informed layer can enforce a narrow but valuable invariant: once an encoded red flag is surfaced, a probabilistic score cannot suppress urgent referral. It cannot guarantee that the extractor surfaces that flag. On identical free-text cases, recall ranged from 56% to 78% to 100% across a naive extractor, MedGemma-4B, and an in-sample tuned reference, respectively. This decomposition—verified rule logic versus measured extraction risk—is the principal contribution. The released contract, benchmark, adapter, metrics, and executable harness make the distinction reproducible and establish a concrete protocol for the required next step: locked evaluation on adjudicated real clinical notes.

Declarations

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Author Contributions (CRediT): Geunsik Lim—Conceptualization, Methodology, Software, Formal analysis, Writing (original draft and editing), and Visualization. The author read and approved the manuscript.

Data and Code Availability: The benchmark, deterministic rules, extractors, evaluation harness, tests, generated tables, and executable notebook are available at <https://github.com/leemgs/stars-audiology>. Model-license acceptance and compatible compute are required to rerun MedGemma. No patient data are used or redistributed.

Ethics: This computational study used no human-subjects data; all cases are author-curated or synthetic. Prospective evaluation on clinical notes will proceed only with the required institutional review and data governance.

AI-Use Disclosure: Generative AI tools assisted with drafting and code scaffolding. The author reviewed and is responsible for all content. No generated text was treated as clinical evidence or independent annotation.

Conflicts of Interest: None declared.

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